

KIRLOSKAR MOTOR HEALTH & DIAGNOSTIC REPORT

AI-Driven Predictive Maintenance & Operational Enhancement Analysis

Dataset Analyzed: kirloskar_continuous_degradation_dataset | Total Telemetry Logs: 10,000

Analysis Timestamp: 2026-07-23 19:36:59

AI Model Accuracy: 99.30%

1. Executive Summary: Motor Condition & Enhancements

OVERALL MOTOR CONDITION: CRITICAL DEGRADATION - URGENT OVERHAUL REQUIRED

Comprehensive analysis of 10,000 telemetry samples from the Kirloskar 1.5 kW motor evaluates overall operational health at 57.3% Healthy, 27.5% Warning, and 15.2% Critical Risk.

The primary root cause of ongoing mechanical degradation is 'Bearing Degradation', accounting for 15.5% of total operating logs. Physical parameter peaks reached 6.05 mm/s vibration (avg: 2.23 mm/s), 106.1 deg C bearing temperature (avg: 65.7 deg C), stator winding temperatures of 106.6 deg C, and current surges up to 4.23 A.

Primary Motor Enhancements & Interventions Summary:

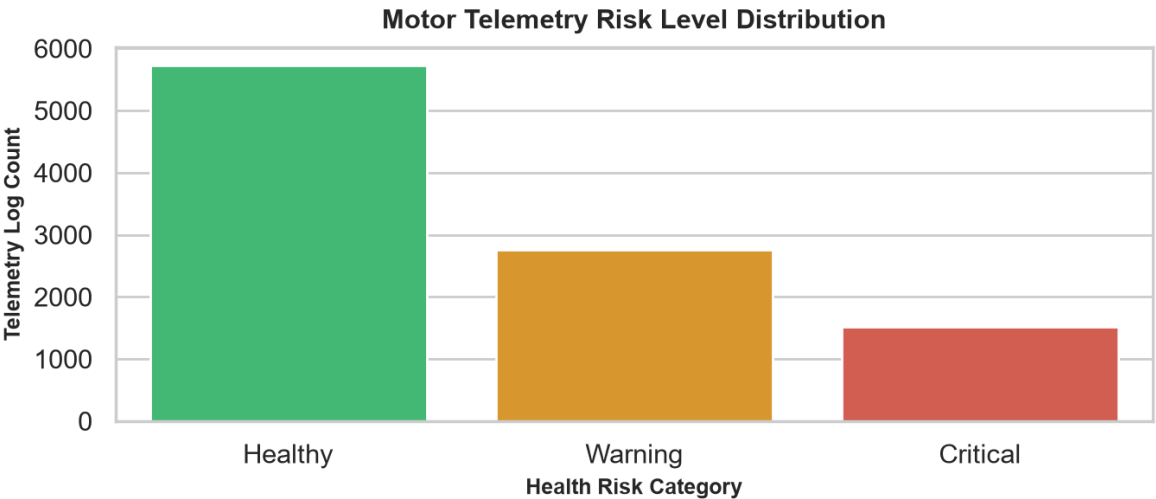
- * Mechanical Enhancements: Replace drive-end bearings, rebalance rotor dynamically to reduce peak vibration from 6.05 mm/s to < 1.5 mm/s, and apply high-temp synthetic grease.
- * Thermal & Electrical Enhancements: Clean cooling fan shrouds to manage stator temperatures (106.6 deg C peak) and adjust VFD soft-starter curves to prevent current spikes above 3.5 A.
- * IoT & AI Model Enhancements: Deploy automated edge alert thresholds for bearing temperature (> 70 deg C) and real-time Random Forest inference for early fault detection.

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2. Telemetry Data Analysis & Risk Distribution Graph

Sensor Parameter	Analyzed Mean	Peak Recorded	95th Percentile	Target Healthy Threshold
Vibration (mm/s)	2.23	6.05	5.20	< 1.80 mm/s
Motor Stator Temp (deg C)	69.3	106.6	99.0	< 65.0 deg C
Bearing Temp (deg C)	65.7	106.1	98.3	< 60.0 deg C
Current Draw (A)	3.24	4.23	3.96	< 3.20 A



Detailed Explanation of the Telemetry Distribution Graph:

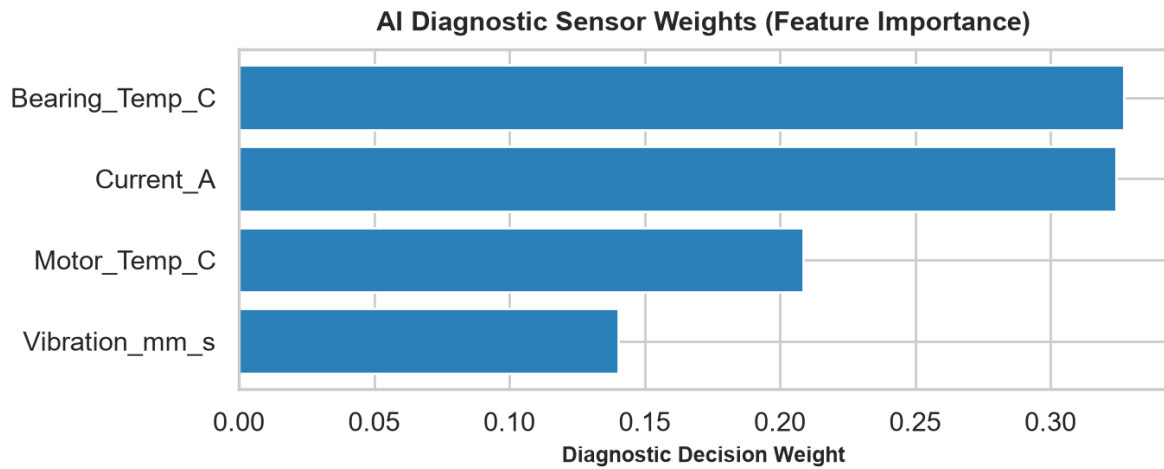
The bar graph above illustrates the distribution of operational risk categories across all 10,000 analyzed telemetry records:

- Healthy (5,727 logs / 57.3%): Baseline operating condition where vibration, temperature, and current remained within optimal limits.
- Warning (2,752 logs / 27.5%): Early-stage degradation where parameters exceeded normal thresholds but remained below critical failure points.
- Critical (1,521 logs / 15.2%): High-risk operational states (e.g. severe bearing failure or heavy overload) requiring urgent maintenance to avoid motor burnout.

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3. AI Feature Prioritization & Sensor Weight Analysis



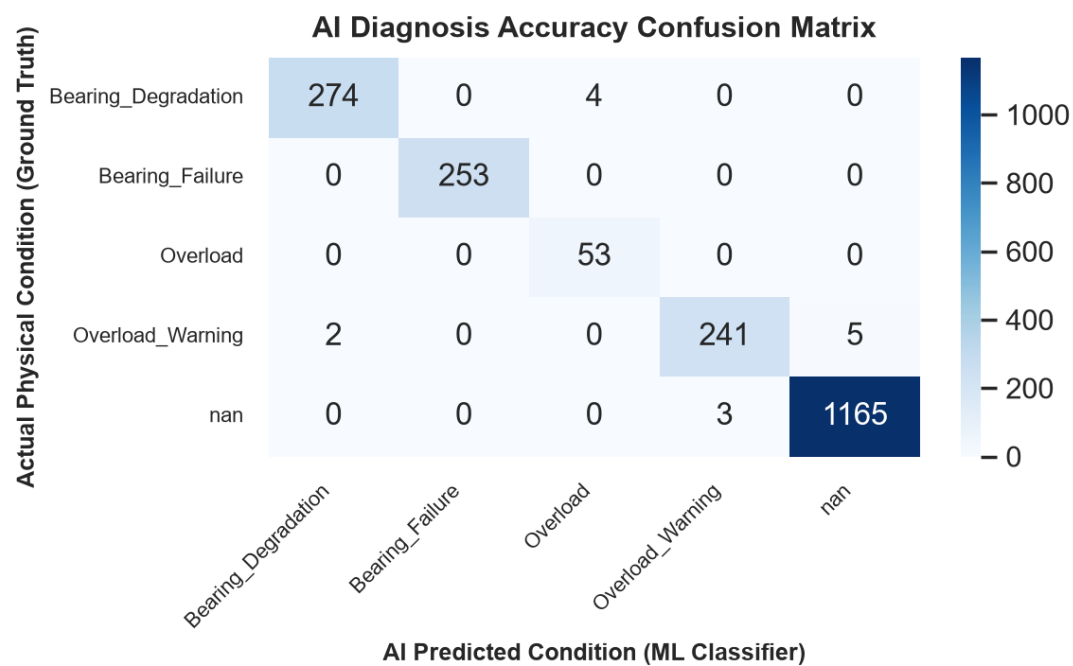
Detailed Explanation of the AI Sensor Weight Graph:

- The horizontal bar chart above displays the relative diagnostic decision weights calculated by the Random Forest model:
- How Weights are Derived:** The model measures the Gini impurity reduction contributed by each sensor feature across 100 decision trees.
 - Primary Leading Indicator:** 'Bearing_Temp_C' holds the highest diagnostic weight (32.7%), meaning it provides the earliest and strongest statistical signal of motor failure.
 - Practical Maintenance Application:** Maintenance personnel should prioritize high-frequency monitoring and sensor calibration for top-weighted features, as anomalies in these channels directly indicate impending physical failure.

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4. AI Diagnostic Accuracy & Confusion Matrix Breakdown



Detailed Explanation of the Confusion Matrix Heatmap:

- The Confusion Matrix evaluates the AI classifier's diagnostic precision on an independent test dataset (2,000 samples):
- Vertical Y-Axis (Actual Physical Condition): Represents the true physical fault category measured by sensor ground truth.
 - Horizontal X-Axis (AI Predicted Condition): Represents the fault category assigned by the Random Forest ML model.
 - Diagonal Cells (Dark Blue): Represent correct AI classifications. High values along the main diagonal confirm 99.30% diagnostic accuracy.
 - Off-Diagonal Cells: Represent misclassifications (False Positives or False Negatives). Minimizing off-diagonal values is critical to prevent false alarms or missed failure events.

Specific Observation: The AI model misclassified 'Overload_Warning' as 'nan' in 5 validation instances (out of 2000 test samples). If the edge monitoring system triggers a 'nan' alert, maintenance personnel should verify whether 'Overload_Warning' symptoms are also present.

Actionable Motor Enhancement & Maintenance Roadmap:

- * **Bearing_Temp_C (Weight: 32.7%):**
Thermal Elevation: Flush bearing housing, apply ISO VG 68 synthetic grease, check seal tightness.
- * **Current_A (Weight: 32.4%):**
Electrical Overload: Inspect mechanical drive binding, verify line voltage, calibrate VFD trip limits.
- * **Motor_Temp_C (Weight: 20.9%):**
Stator Overheating: Clear cooling fins, inspect external fan blades, verify voltage symmetry.
- * **Vibration_mm_s (Weight: 14.0%):**
High Oscillation: Replace worn bearings, execute laser coupling alignment, and rebalance rotor.